

MOHIT CHANDANA

Senior Full-Stack Engineer & Team Lead | Rails · Vue.js · AI/LLM Integration | HIPAA Healthcare Systems
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Washington, D.C. (Remote-Open)

PROFESSIONAL SUMMARY

Senior Rails Engineer and Team Lead with 12+ years shipping enterprise-grade, HIPAA-compliant healthcare systems. Specialized in AI-powered backend architecture — building local LLM (Ollama) and RAG pipelines that keep patient data on-premise. Proven track record across pharmaceutical, health-tech, insurance, and SaaS domains with 5 personal projects live in production. Expert across the modern Rails stack: Rails 8, Vue.js 3, PostgreSQL/pgvector, GraphQL, Sidekiq, and AWS.

TECHNICAL SKILLS

Backend: Ruby on Rails 8.x/7.x, Ruby, Sinatra, RESTful APIs, GraphQL, Python, Django

Frontend: Vue.js 3 (Composition API), Vite.js, React.js, JavaScript/ES6+, Tailwind CSS, Pinia

AI & ML: Ollama (local LLM), RAG pipelines, pgvector, OpenAI GPT-4o, Google Gemini, Anthropic Claude API

Databases: PostgreSQL, pgvector, MySQL, SQLite, MongoDB, Redis

Cloud & DevOps: AWS (EC2, S3), Docker, GitHub Actions CI/CD, Cloudflare R2, Nginx, Render

Testing: RSpec (951+ examples), Cypress, Vitest, Factory Bot, Capybara

Compliance: HIPAA, PHI handling, field-level encryption (Lockbox), audit logging, Rack::Attack, 2FA/TOTP

Other: Twilio API, ActionCable WebSockets, Sidekiq, AASM, acts_as_tenant, MCP Protocol, Go

PROFESSIONAL EXPERIENCE

Vanda Pharmaceuticals — Washington, D.C.

Dec 2018 – Present

Senior Ruby on Rails Engineer / Team Lead

- Architected and led development of Ultra CRM, a HIPAA-compliant customer relationship management system serving the full pharmaceutical commercial operations lifecycle — built on Rails 8 API backend, Vue.js 3 SPA frontend, PostgreSQL, and GraphQL.
- Led a 3-engineer frontend team building Ultra Vue, a Vue.js 3 + Vite.js SPA with Tailwind CSS; migrated legacy static HTML/CSS/JS applications to Vue SSR, cutting build times by 60% and improving developer iteration speed significantly.
- Developed RESTful and GraphQL APIs handling real-time CRM data updates across 40k+ patient records, with Redis caching and Sidekiq background jobs for asynchronous processing.
- Integrated Twilio API for SMS and call functionality within CRM, enabling compliant patient communication workflows with full audit logging for HIPAA adherence.
- Optimized critical PostgreSQL queries using composite indexing, eager loading, and query analysis — reducing average response times from 2.1s to under 200ms across high-traffic endpoints.
- Established RSpec and Cypress testing standards for the team; wrote comprehensive unit and E2E test suites, maintaining high coverage across all CRM modules.
- Conducted weekly code reviews, led Agile sprint planning, and mentored 2 junior engineers on Rails best practices, service object patterns, and compliance-first development.

UnitedHealth Group / Optum — Broomfield, CO

May 2017 – Nov 2018

Senior Ruby on Rails Engineer / Architect

- Built SMS Manager — a high-throughput healthcare communication platform using Rails, Sidekiq, and Twilio API, processing asynchronous SMS messaging between patients and providers.
- Designed PostgreSQL database models with optimized indexing strategies for message queuing and delivery tracking, supporting compliance with healthcare communication regulations.

- Led Agile stand-ups and sprint planning as technical lead; mentored junior developers on RESTful API design, background job architecture, and RSpec testing patterns.
- Implemented CI/CD deployment pipelines for continuous integration and zero-downtime deploys across staging and production environments.

Apple, Inc. / NetApp — Austin, TX

Jun 2016 – May 2017

Senior Ruby on Rails Engineer

- Built StorageAdmin — a web application for managing NetApp enterprise storage devices, using Rails API and Sinatra microservices for device configuration and monitoring operations.
- Implemented LDAP authentication for enterprise SSO, integrated Resque for async background processing, and used RABL for efficient JSON serialization across storage management APIs.
- Designed responsive UI components using HAML, SCSS, and jQuery DataTables for large-dataset visualization; performed PostgreSQL migrations and performance optimizations.
- Collaborated cross-functionally with Apple infrastructure and NetApp storage teams in Agile sprint cycles, translating complex enterprise storage requirements into clean Ruby API abstractions and maintainable service-layer code.

Multi-Tech Systems, Inc. — Mounds View, MN

Jan 2015 – May 2016

Ruby on Rails Developer

- Developed DeviceHQ — a cloud-based IoT device management and monitoring SPA using Backbone.js frontend and Rails backend, managing thousands of M2M/IoT connected devices.
- Built RESTful APIs for IoT device configuration and monitoring; managed MongoDB collections and integrated AWS S3 for scalable cloud storage and deployment.
- Developed authentication modules with role-based access control and implemented automated testing using RSpec and Capybara — maintaining reliability across a platform managing thousands of live connected devices.

PERSONAL PROJECTS — 5 LIVE PRODUCTION APPS

RFE Ready rfeready.com

Stack: *Rails 8 · Vue.js 3 · PostgreSQL · pgvector · OpenAI GPT-4o · ActionCable · Docker · AWS EC2 | 2025 – Present*

- Architected production-grade RAG pipeline integrating OpenAI GPT-4o with PostgreSQL pgvector (1,536-dimensional embeddings) for semantic RFE knowledge search — reducing legal draft preparation from 4+ hours to 15 minutes with 95% attorney approval rate, processing 20+ immigration cases/day.
- Built Google Docs-style real-time collaborative editing using ActionCable WebSockets with cursor synchronization, presence indicators, and optimistic locking — supporting 100+ concurrent connections at <50ms broadcast latency across 3+ simultaneous attorneys.
- Engineered zero-trust multi-tenant SaaS architecture (20 tables, row-level data isolation) with JWT token rotation, TOTP-based 2FA, AES-256-GCM PII encryption, and comprehensive audit logging — achieving 100% tenant isolation over 6 months in production.
- Delivered 40,000 lines of production code with 90.7% test coverage (951 RSpec examples + 451 Vitest unit tests), Brakeman security scanning, and GitHub Actions CI/CD with zero-downtime Docker deployments — zero critical bugs in 6 months.

LullaTrack lullatrack.com

Stack: *Rails 8 · Vue.js 3 · Google Gemini 2.5 Flash · Twilio SMS · Telegram Bot · Gmail IMAP · AWS EC2 · Docker · Sidekiq | 2025 – Present*

- Integrated Google Gemini 2.5 Flash as a natural-language parser across 3 inbound channels (Twilio SMS, Telegram Bot, Gmail IMAP polling) — enabling parents to log 7 record types via free-text messages with AI entity extraction, multi-model fallback, and smart deduplication (± 2 -min time windows).
- Engineered WHO growth percentile system using LMS (Lambda-Mu-Sigma) statistical method with Z-score interpolation across 5 percentile curves; built Prawn-based PDF report generator producing 8-section doctor visit summaries with feeding analytics and vaccination schedules.
- Architected full-stack platform with 14 data models, 128 API endpoints, and 20 views covering feedings, diapers, sleep, growth, vaccinations, milestones, and appointments across multiple family caregivers.

- Deployed on AWS EC2 with Docker multi-stage builds, GitHub Actions CI/CD, Cloudflare R2 photo storage, Sidekiq with 5 scheduled background jobs, and Rack::Attack rate limiting (4 throttle tiers).

Memvra memvra.com

Stack: Go · SQLite · sqlite-vec · Model Context Protocol (MCP) · Multi-LLM (Claude · OpenAI · Gemini · Ollama) | `brew install memvra/tap/memvra` | 2025 – Present

- Implemented MCP (Model Context Protocol) server integrating with Claude Code, Cursor, and Windsurf IDEs — exposing 8 tools for context injection, session persistence, and cross-tool state synchronization across 30+ programming languages.
- Architected multi-provider LLM abstraction layer supporting Claude, OpenAI, Gemini, and Ollama with unified streaming interface and adapter pattern — enabling seamless model switching across 15,000+ LOC Go codebase.
- Engineered hybrid conflict detection combining vector similarity search (SQLite + sqlite-vec, 768-dimensional embeddings) with LLM semantic analysis — achieving 70–90% cache hit rate via LRU embedding cache with two-stage false-positive filtering.
- Optimized database performance 5.6x (45s → 8s indexing) via SQLite WAL mode tuning, 5-connection pooling, prepared statement caching for vector ops, and batch transaction processing for bulk embedding inserts.

Asclepius AI

Stack: Rails 7.2 · Vue.js 3 · Ollama (local LLM) · RAG · PostgreSQL · Redis · Sidekiq · AWS S3 · Lockbox | 2024 – Present

- Built HIPAA-compliant AI document management system for pharmaceutical organizations using local Ollama LLM with RAG — keeping all PHI on-premise without any data leaving the infrastructure; implemented Lockbox field-level encryption, role-based access control (Admin/Doctor/Researcher/Auditor), and comprehensive PHI audit trails.
- Designed end-to-end semantic search and Q&A pipeline: PDF/DOCX text extraction → chunk splitting → embedding generation → pgvector storage → context-injected LLM response; all processing via Sidekiq background jobs with real-time monitoring dashboard for job status and system health.
- Implemented HIPAA security controls including Redis-backed session management (15-minute inactivity timeout), breach notification procedures, security risk assessments, and Business Associate Agreement (BAA) compliance documentation.
- Integrated AWS S3 with server-side encryption and versioning for HIPAA-compliant cloud document storage; built batch upload system supporting PDF, DOCX, and image formats with drag-and-drop interface and Sidekiq-powered background processing.

Massklan www.massklan.com

Stack: Rails · PostgreSQL · Devise · Active Storage · Cloudinary | 2023 – Present

- Built production tournament management platform for badminton with dynamic bracket generation (Round of 32/16 through Finals), ELO-based skill rating system with win/loss/opponent-strength weighting, consolation bracket logic for double elimination, and division management across 5 competition categories (Men's/Women's Singles & Doubles, Mixed).
- Implemented partner request system allowing players to find and invite doubles partners, match scoring with best-of-3 game sets and automatic winner advancement, walkover/bye handling, and bronze medal match logic for double elimination tournaments.
- Optimized PostgreSQL queries for complex tournament bracket traversal and player statistics — built rating history tracking showing player progression over time with detailed per-match analytics and leaderboards.
- Designed admin dashboard for tournament organizers to manage divisions, schedule matches, update results, and oversee player registrations with deadline management, withdrawal workflows, and Cloudinary-powered image uploads.

EDUCATION

Master's in Computer Science — University of Central Missouri

2014

Bachelor's in Computer Science Engineering — JNTU

2013